

LAB EXPERIMENT

ITA0506-COMPUTER VISION

EXPNO:20

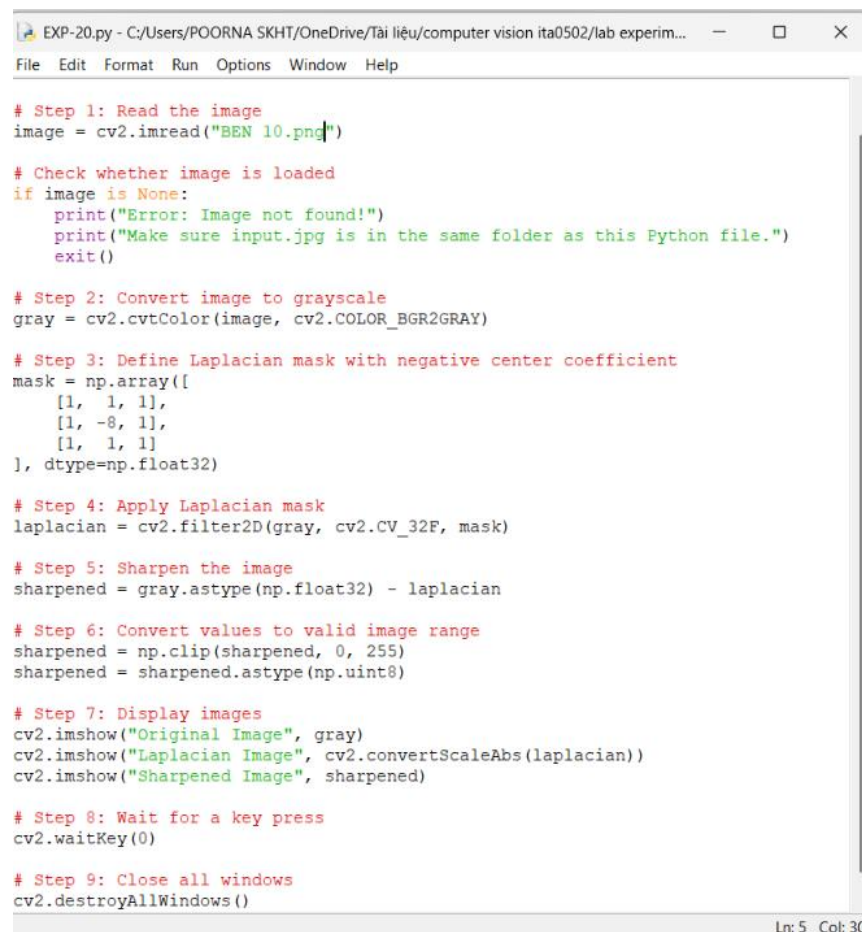
Aim

To sharpen an image using a Laplacian mask with a negative center coefficient.

Algorithm

1. Read the input image.
2. Convert the image to grayscale.
3. Define a Laplacian sharpening mask with a negative center coefficient.
4. Apply the mask using convolution.
5. Obtain the sharpened image.
6. Display the original and sharpened images.

Code:



```
EXP-20.py - C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experim...
File Edit Format Run Options Window Help

# Step 1: Read the image
image = cv2.imread("BEN 10.png")

# Check whether image is loaded
if image is None:
    print("Error: Image not found!")
    print("Make sure input.jpg is in the same folder as this Python file.")
    exit()

# Step 2: Convert image to grayscale
gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

# Step 3: Define Laplacian mask with negative center coefficient
mask = np.array([
    [1, 1, 1],
    [1, -8, 1],
    [1, 1, 1]
], dtype=np.float32)

# Step 4: Apply Laplacian mask
laplacian = cv2.filter2D(gray, cv2.CV_32F, mask)

# Step 5: Sharpen the image
sharpened = gray.astype(np.float32) - laplacian

# Step 6: Convert values to valid image range
sharpened = np.clip(sharpened, 0, 255)
sharpened = sharpened.astype(np.uint8)

# Step 7: Display images
cv2.imshow("Original Image", gray)
cv2.imshow("Laplacian Image", cv2.convertScaleAbs(laplacian))
cv2.imshow("Sharpened Image", sharpened)

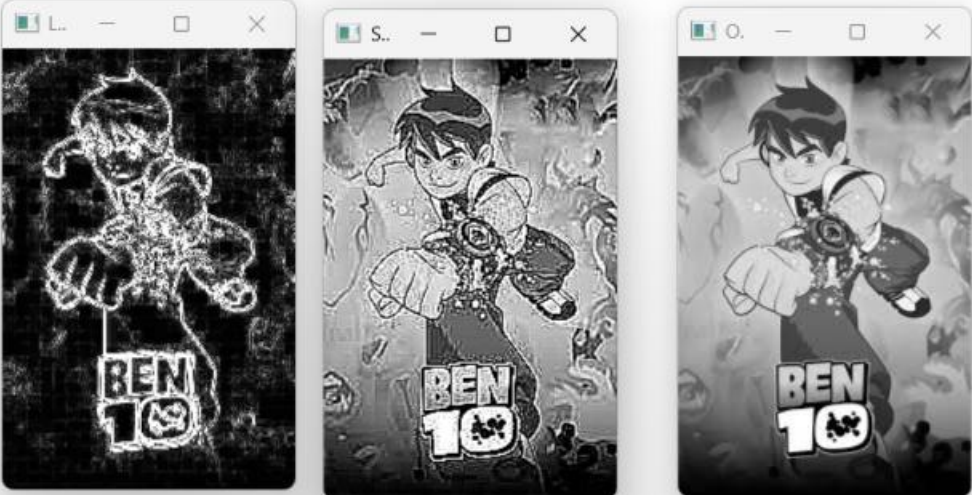
# Step 8: Wait for a key press
cv2.waitKey(0)

# Step 9: Close all windows
cv2.destroyAllWindows()

Ln: 5 Col: 30
```

OUTPUT :

```
*IDLE Shell 3.13.3*
File Edit Shell Debug Options Window Help
Python 3.13.3 (tags/v3.13.3:6280bb5, Apr 8 2025, 14:47:33) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experiments/EXP-19.py
= RESTART: C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experiments/EXP-20.py
```



Ln: 7 Col: 0

Result

The input image is successfully sharpened using a Laplacian mask with a negative center coefficient. The edges and fine details become more prominent in the sharpened image.